DATA ANALYTICS PROJECT REPORT

Customer Shopping Behaviour Analysis

Understanding how customers shop to improve sales, satisfaction and long-term loyalty

Dataset: Customer shopping behaviour data (3,900 transactions)

Deliverables: Analytical findings and interactive Power BI dashboard

Report date: 18 October 2026

Table of Contents

1. Executive Summary..... 3
2. Business Problem.....4
3. Project Objectives.....5
4. Our Approach..... 6
5. Dataset Overview..... 7
6. Analysis and Key Findings.....8
7. Interactive Power BI Dashboard..... 14
8. Summary of Key Findings.....16
9. Recommendations..... 17
10. Conclusion.....18
11. Limitations and Next Steps.....19

1. Executive Summary

A leading retail company wants to understand how its customers shop so that it can grow sales, raise customer satisfaction and build long-term loyalty. Management has observed shifting purchase patterns across demographics, product categories and sales channels, and wants to know which factors (discounts, reviews, seasons and payment preferences) influence purchase decisions and repeat buying.

This report presents the results of an analysis of **3,900 customer transactions**, together with an interactive Power BI dashboard that lets stakeholders explore the same insights on demand. Nine business questions were answered, covering revenue drivers, discounts, subscriptions, shipping, product performance and customer loyalty.

Headline findings

- **Revenue is male-driven by volume.** Male customers generated $157,890 (67.7%) of the $233,081 total revenue versus $75,191 (32.3%) from female customers, yet average spend per transaction is almost identical ($59.54 male vs $60.25 female).
- **Discounts do not shrink baskets.** 839 customers used a discount and still spent above the overall average of $59.76, with an average spend of $79.79.
- **Subscribers do not spend more per purchase.** Subscribers average $59.49 per transaction versus $59.87 for non-subscribers, and only 27% of customers are subscribed.
- **The customer base is mature.** Only 83 customers (2.1%) are new, while 1,935 are Loyal and 1,882 are Returning, which points to an acquisition gap rather than a retention problem.
- **Express shipping customers spend slightly more** ($60.48 vs $58.46 for Standard), about 3.5% higher.
- **Repeat buyers are only marginally more likely to subscribe** (27.6% compared with 22.4% for other customers), indicating untapped subscription potential.

The report closes with recommendations on marketing, promotions, subscriptions and customer acquisition.

2. Business Problem

2.1 Background

A leading retail company wants to better understand its customers' shopping behaviour in order to improve sales, customer satisfaction and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories and sales channels (online versus offline).

2.2 Problem Statement

Management is particularly interested in uncovering which factors, such as discounts, reviews, seasons or payment preferences, drive customer decisions and repeat purchases. Without a structured analysis of consumer behaviour, the company cannot confidently target its marketing spend, tune its promotions or design loyalty programmes.

2.3 Overarching Business Question

> **Core question**
> How can the company leverage consumer shopping data to identify trends, improve customer engagement and optimise marketing and product strategies?

2.4 Business Challenges Addressed

- Understanding who the customers are and how much each group contributes to revenue.
- Measuring the effect of discounts and promotions on purchase value.
- Evaluating whether subscription, shipping choice and purchase history are linked to higher spending or loyalty.
- Identifying best-performing and most-discounted products to support inventory and promotion planning.
- Segmenting customers by purchase history so that marketing can be targeted.

3. Project Objectives

3.1 Primary Objective

Convert raw customer shopping data into actionable insight that guides marketing, product and customer-engagement decisions.

3.2 Supporting Objectives

- Prepare a reliable, analysis-ready dataset.
- Answer nine specific business questions about customer behaviour.
- Deliver an interactive dashboard for ongoing monitoring by stakeholders.
- Provide clear recommendations that can be acted upon.

3.3 Business Questions Analysed

Table 1: Business questions answered

#	Business question
Q1	What is the total revenue generated by male versus female customers?
Q2	Which customers used a discount but still spent more than the average purchase amount?
Q3	Which are the top 5 products with the highest average review rating?
Q4	How does the average purchase amount compare between Standard and Express shipping?
Q5	Do subscribed customers spend more than non-subscribers?
Q6	Which 5 products have the highest percentage of purchases with a discount applied?
Q7	How can customers be segmented into New, Returning and Loyal groups, and how large is each?
Q8	What are the top 3 most purchased products within each category?
Q9	Are repeat buyers (more than 5 previous purchases) also likely to subscribe?

4. Our Approach

The project followed a simple five-step process, moving from understanding the business need to delivering insight and recommendations.



Figure 1: Project approach at a glance

Table 2: Approach summary

Step	Stage	What was done
1	Business problem	Defined the problem, the core business question and the objectives with the stakeholders
2	Data collection and preparation	Collected the shopping data, checked its quality and prepared it for analysis
3	Analysis and insight generation	Analysed the data to answer the nine business questions
4	Interactive dashboard	Built a Power BI dashboard so that findings can be explored on demand
5	Findings and recommendations	Summarised the results and translated them into recommended actions

Data quality was treated carefully: the dataset was checked for duplicates and gaps, and the small number of missing review ratings (under 1% of records) was treated so that no transactions had to be excluded. Customer age bands and purchase-frequency measures were added to support deeper analysis. The analysis was performed with Python and SQL, and results were visualised in Power BI.

5. Dataset Overview

The dataset holds **3,900 customer transactions**, one per customer, described by 18 attributes. These fall into five groups:

- **Customer profile:** age, gender and location (50 US states).
- **Product:** item purchased (25 products), category, size and colour.
- **Purchase:** purchase amount, season, payment method and shipping type.
- **Engagement:** review rating, subscription status, discount applied, previous purchases and purchase frequency.

5.1 Key Characteristics

Table 3: Dataset at a glance

Measure	Value
Total transactions / customers	3,900
Total revenue	$233,081
Average purchase amount	$59.76 (range $20 to $100)
Average review rating	3.75 out of 5
Customer age	18 to 70 years (average 44)
Gender split	Male 68% (2,652), Female 32% (1,248)
Category mix	Clothing 44.5%, Accessories 31.8%, Footwear 15.4%, Outerwear 8.3%
Subscribers	27.0% of customers
Transactions with a discount	43.0%

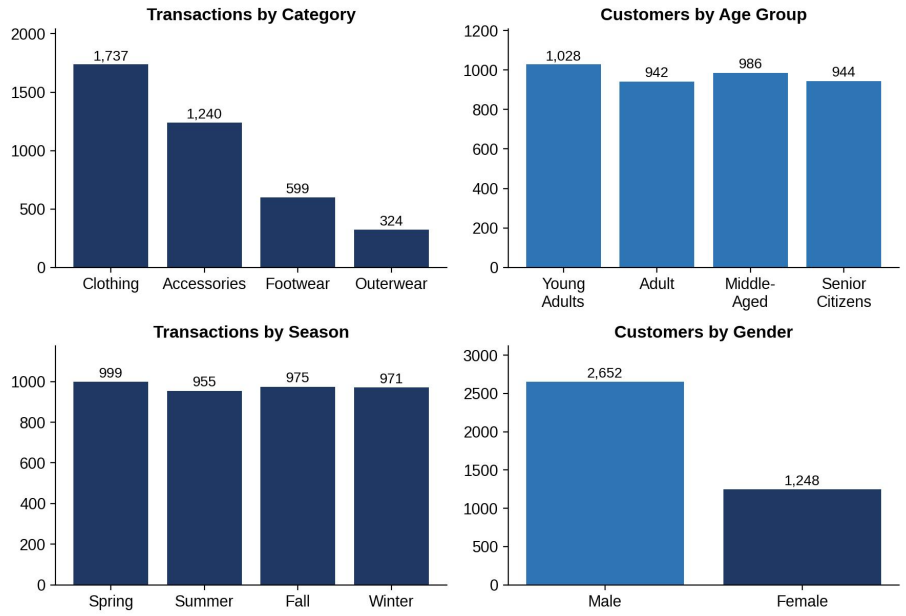


Figure 2: Distribution of transactions by category, age group, season and gender

6. Analysis and Key Findings

This section presents the answer to each business question, followed by the business interpretation.

6.1 Q1: Revenue by Gender

Business question: What is the total revenue generated by male versus female customers?

Table 4: Revenue by gender

Gender	Customers	Total revenue	Share of revenue	Avg purchase
Male	2,652	$157,890	67.7%	$59.54
Female	1,248	$75,191	32.3%	$60.25

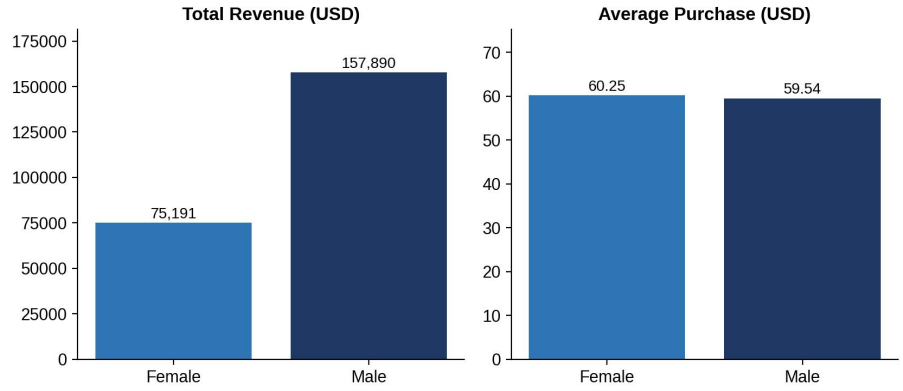


Figure 3: Revenue and average purchase by gender

> **Insight**
> Male customers contribute more than twice the revenue of female customers, but only because there are about 2.1 times as many of them. Female customers spend slightly more per transaction, so the gap is a customer-volume issue, not a spending issue. Growing the female customer base is a clear revenue opportunity.

6.2 Q2: Discount Users Who Still Spent Above Average

Business question: Which customers used a discount but still spent more than the average purchase amount?

Compared with the overall average purchase of **$59.76**, **839 customers** used a discount and still spent more. Their average purchase was **$79.79**, about 33.5% above the benchmark, and they represent roughly half (50.0%) of the 1,677 discounted transactions.

> **Insight**
> Discounts are not only attracting low-value baskets. Average spend with a discount ($59.28) is nearly identical to spend without one ($60.13), and half of discount users spend above average. Discounts appear to be a low-risk lever for conversion, and these 839 customers are a strong target group for loyalty offers.

6.3 Q3: Top 5 Products by Average Review Rating

Business question: Which are the top 5 products with the highest average review rating?

Table 5: Top 5 products by average rating

Rank	Product	Average rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	T-shirt	3.78

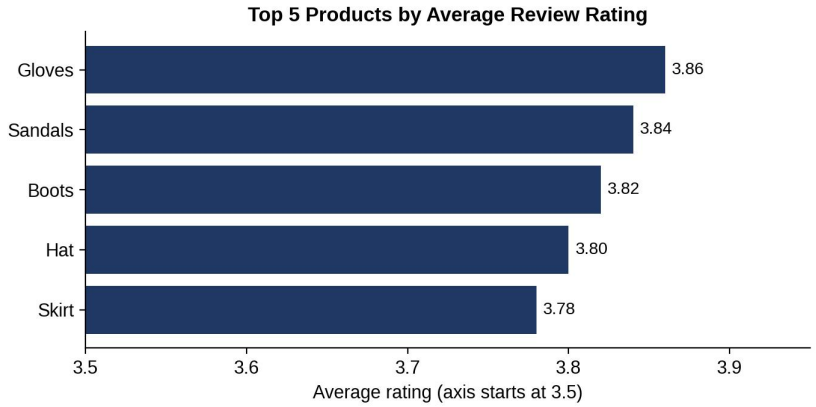


Figure 4: Top 5 products by average review rating

> **Insight**
> The gap between first and fifth place is only 0.08 points and all are close to the overall average of 3.75, so satisfaction is consistent across the range. Gloves, Sandals and Boots can be highlighted in marketing as customer favourites, while the overall rating level suggests room to improve quality or experience across the catalogue.

6.4 Q4: Standard versus Express Shipping

Business question: How does the average purchase amount compare between Standard and Express shipping?

Table 6: Average purchase by shipping type

Shipping type	Transactions	Average purchase
Express	646	$60.48
Standard	654	$58.46

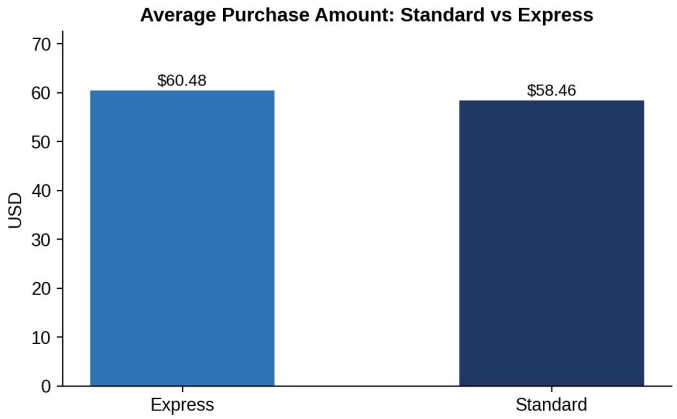


Figure 5: Average purchase amount: Standard versus Express shipping

Insight
Express customers spend $2.02 more per order (about 3.5%) than Standard customers. The difference is modest, but suggests that customers who pay for speed are slightly more committed buyers. Express could be promoted at checkout or used as an incentive above a basket threshold.

6.5 Q5: Do Subscribed Customers Spend More?

Business question: Compare average spend and total revenue between subscribers and non-subscribers.

Table 7: Spend by subscription status

Group	Customers	Average spend	Total revenue
Subscribers	1,053	$59.49	$62,645
Non-subscribers	2,847	$59.87	$170,436

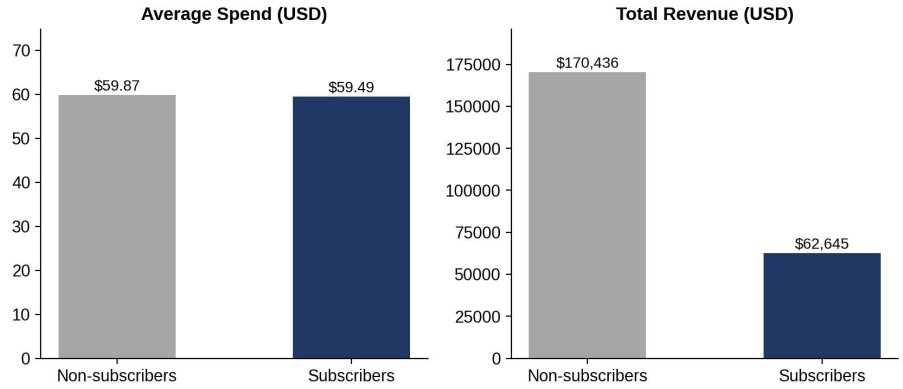


Figure 6: Average spend and total revenue by subscription status

Insight
Subscribers do not spend more per purchase. Their average spend is marginally lower, and they generate 26.9% of revenue, in line with their 27% share of customers. The subscription programme should therefore be judged on retention and purchase frequency, and its benefits may need to be redesigned to lift spend.

6.6 Q6: Products with the Highest Discount Share

Business question: Which 5 products have the highest percentage of purchases with a discount applied?

Table 8: Top 5 products by share of discounted purchases

Rank	Product	Purchases with discount
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%
5	Pants	47.37%

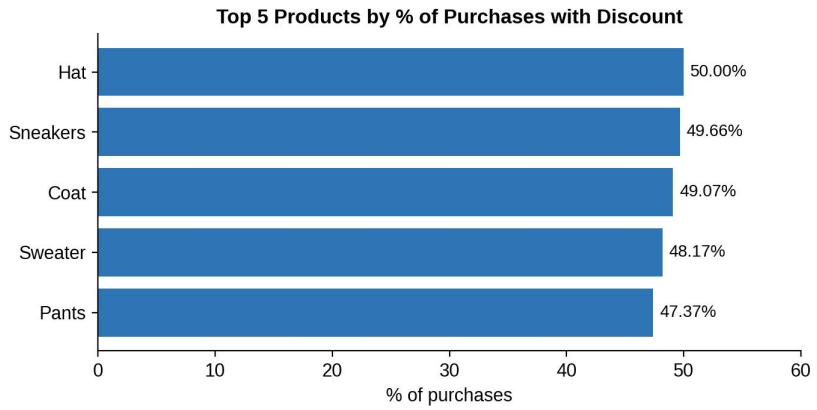


Figure 7: Top 5 products by percentage of purchases with a discount

> **Insight**
> Discounts apply to 43% of all transactions, so these five products are discounted 4 to 7 percentage points more often than average. Management should confirm that these products do not depend on discounts to sell, and review the margin impact of discounting Coats and Sweaters.

6.7 Q7: Customer Segmentation

Business question: Segment customers into New, Returning and Loyal groups based on their number of previous purchases.

Table 9: Customer segments

Segment	Previous purchases	Customers	Share
New	1 or fewer	83	2.1%
Returning	2 to 25	1,882	48.3%
Loyal	26 to 50	1,935	49.6%
VIP	More than 50	0	0%

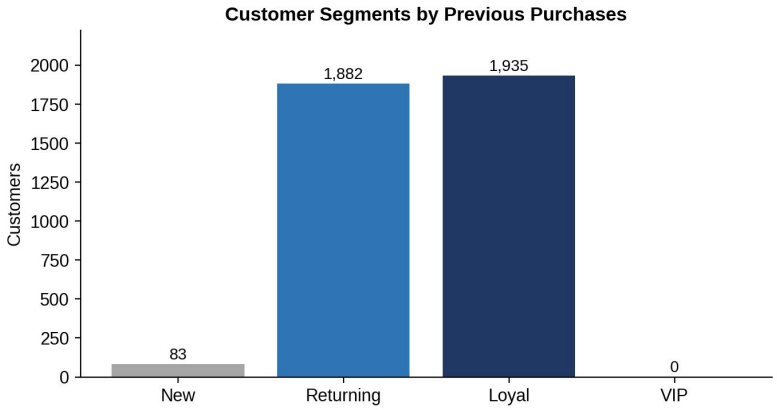


Figure 8: Customer segments by number of previous purchases

Insight
Almost the entire customer base (97.9%) has made at least two earlier purchases, and nearly half are Loyal. Very few new customers are entering the business, which indicates an acquisition challenge. No customer qualifies as VIP because no customer has more than 50 previous purchases, so the VIP threshold should be recalibrated (for example 40 or more) if a top tier is wanted.

6.8 Q8: Top 3 Products within Each Category

Business question: What are the top 3 most purchased products within each category?

Table 10: Top products per category (number of purchases)

Category	Rank 1	Rank 2	Rank 3
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Pants (171)	Blouse (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	n/a (only 2 products)

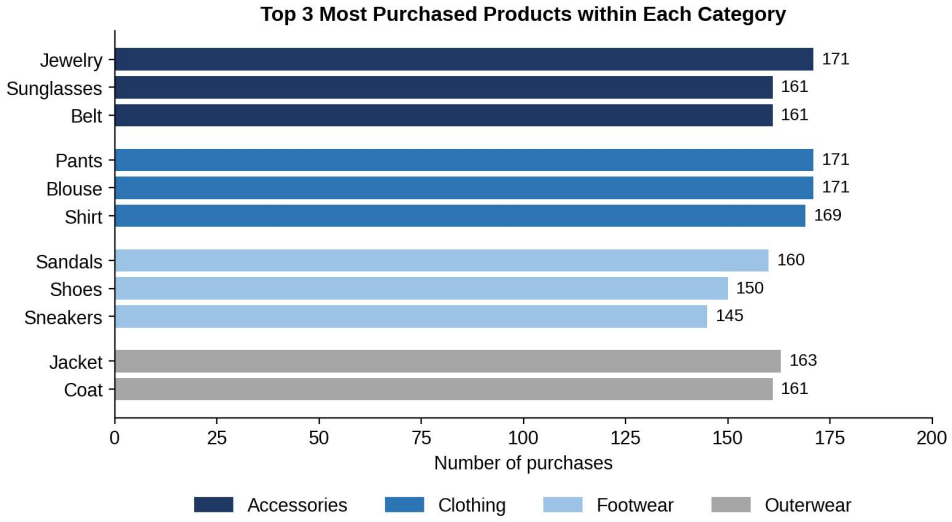


Figure 9: Top 3 most purchased products within each category

Insight
Demand is evenly spread within each category, with counts differing by only a few orders, so no single hero product dominates. Outerwear has only two products, a narrow range compared with the other categories. Where counts tie (Pants and Blouse; Sunglasses and Belt), the order shown is not meaningful.

6.9 Q9: Repeat Buyers and Subscription

Business question: Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

Table 11: Subscription status of repeat buyers

Group	Subscribed	Not subscribed	Subscription rate
Repeat buyers (more than 5)	958	2,518	27.6%
5 or fewer purchases	95	329	22.4%
All customers	1,053	2,847	27.0%

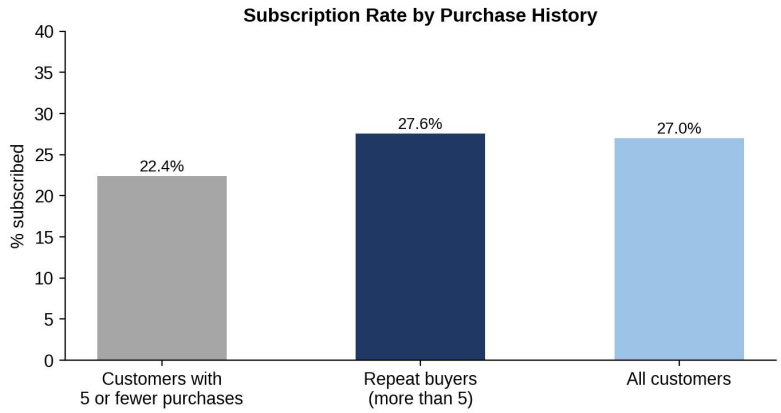


Figure 10: Subscription rate by purchase history

Insight
There are 3,476 repeat buyers, but only 958 (27.6%) subscribe, just 5 percentage points above less frequent buyers. Purchase loyalty is not translating into subscription uptake, and roughly 2,500 unsubscribed repeat buyers are the most obvious target for a subscription campaign.

7. Interactive Power BI Dashboard

To make the findings accessible to business users, an interactive **Customer Behavior Analytics Dashboard** was built in Power BI. It presents the key measures on a single page and lets stakeholders filter the view without technical assistance.

7.1 Dashboard Components

Table 12: Dashboard contents

Component	What it shows
Headline KPI cards	Number of customers (3,900), average purchase amount ($59.76) and average review rating (3.75)
Subscription Status (donut chart)	Number of customers who are subscribed versus not subscribed
Revenue by Category (column chart)	Revenue contribution of Clothing, Accessories, Footwear and Outerwear
Revenue Across Shipping Types (pie chart)	Share of revenue by shipping option
Revenue Across Age Group (column chart)	Revenue by customer age band
Revenue Across Payment Methods (line chart)	Revenue by payment method
Filters (slicers)	Gender, Subscribers, Categories and Shipping Type; all charts update together

7.2 Dashboard Overview



Figure 11: Customer Behavior Analytics Dashboard (Power BI)

7.3 What the Dashboard Reveals

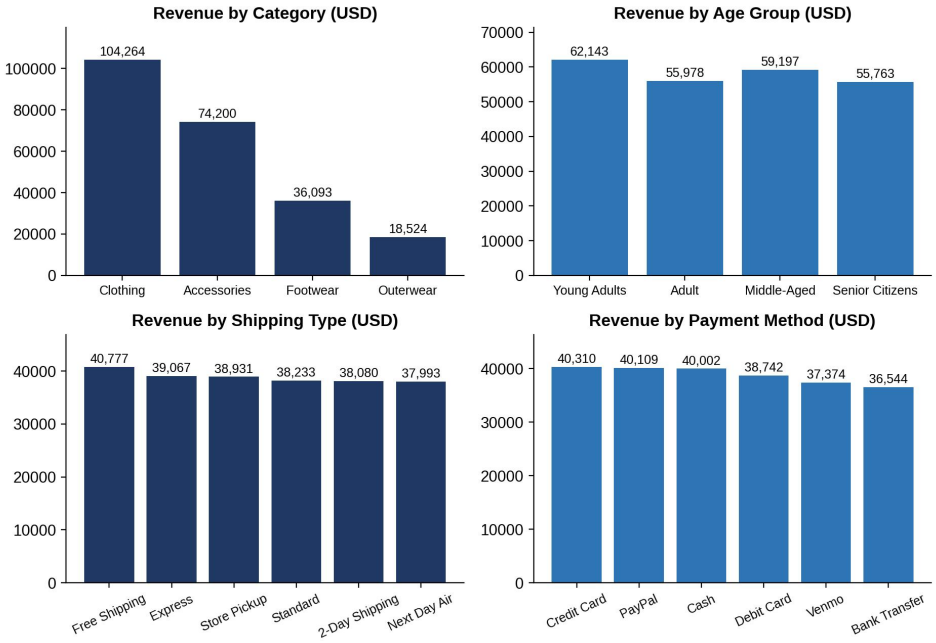


Figure 12: Revenue breakdowns presented in the dashboard

- **Category:** Clothing (44.7%) and Accessories (31.8%) generate over three-quarters of revenue; Outerwear contributes only 7.9%.
- **Age group:** revenue is spread evenly, from 26.7% (Young Adults) to 23.9% (Senior Citizens), so no age band is dominant.
- **Shipping:** all six shipping options earn between 16.3% and 17.5% of revenue; Free Shipping is the largest.
- **Payment method:** Credit Card, PayPal and Cash lead (about 17% each), while Bank Transfer is lowest at 15.7%.

7.4 How Stakeholders Can Use It

- Select a gender, category, subscription status or shipping type to see how every KPI and chart changes.
- Monitor customer count, average spend and satisfaction at a glance.
- Spot under-performing categories or customer groups for follow-up.

8. Summary of Key Findings

Table 13: Findings summary

Analysis	Key result	Business meaning
Q1 Gender	Male $157,890 vs Female $75,191	Gap is driven by customer volume, not spend per order
Q2 Discounts	839 discount users spent above $59.76 (avg $79.79)	Discounts do not reduce basket value
Q3 Ratings	Gloves (3.86), Sandals (3.84), Boots (3.82)	Satisfaction is consistent; narrow spread
Q4 Shipping	Express $60.48 vs Standard $58.46	Express buyers spend about 3.5% more
Q5 Subscription	Subscribers $59.49 vs non-subscribers $59.87	Subscription does not raise spend per order
Q6 Discount share	Hat 50.00%, Sneakers 49.66%, Coat 49.07%	Promotions are concentrated on a few products
Q7 Segments	New 83, Returning 1,882, Loyal 1,935	Mature base; weak new-customer inflow
Q8 Top products	Jewelry, Pants, Blouse lead (171 orders each)	No single dominant product per category
Q9 Repeat buyers	27.6% of repeat buyers subscribe	About 2,500 repeat buyers untapped for subscription

9. Recommendations

The following actions follow directly from the findings. Several differences in the data are small, so each recommendation should be tested on a limited scale before wider rollout.

9.1 Marketing and Customer Acquisition

- **Acquire new customers.** With only 2.1% of customers classified as New, invest in acquisition campaigns and referral incentives.
- **Grow the female customer segment.** Female customers already spend slightly more per order, so targeted campaigns and assortments could close the volume gap.

9.2 Promotion and Pricing Strategy

- **Keep discounts, but target them.** Use discounts to convert hesitant buyers and offer exclusive deals to the 839 high-spending discount users.
- **Review discount dependence** on Hat, Sneakers, Coat, Sweater and Pants, and check margins before extending promotions.

9.3 Subscription and Loyalty

- **Launch a subscription campaign** aimed at the roughly 2,500 repeat buyers who are not yet subscribed.
- **Redesign subscriber benefits** (for example free or Express shipping, early access, bundles) so that subscription lifts spend per order.
- **Recalibrate the VIP tier** so that the top customers receive recognition and rewards.

9.4 Product and Fulfilment

- **Promote best-rated products** such as Gloves, Sandals and Boots.
- **Encourage Express shipping** at checkout, since Express customers spend more.
- **Review the Outerwear range,** which has only two products and the lowest average spend ($57.17) of the four categories.

10. Conclusion

This project turned raw customer shopping data into structured, decision-ready insight, supported by an interactive Power BI dashboard for ongoing monitoring.

The analysis shows a customer base that is loyal and mature but narrow: revenue is dominated by male customers because of volume, discounts and subscriptions do not meaningfully change spend per transaction, and very few new customers are joining. The most promising opportunities are to acquire new customers, widen the female customer base, convert the large pool of unsubscribed repeat buyers into subscribers, and redesign subscription benefits so that they add measurable value.

Together, these findings answer the original business question: the company can use its shopping data to target acquisition, tune promotions, strengthen loyalty programmes and prioritise products.

11. Limitations and Next Steps

11.1 Limitations

- **Small differences:** many gaps between groups (for example $59.49 vs $59.87 for subscribers) are small, and no statistical significance tests were performed.
- **No time dimension:** the data has no transaction date, so trends over time, retention and customer lifetime value cannot be measured.
- **No channel or margin data:** online versus offline channels, product cost and profit are not included, so channel performance and discount profitability cannot be assessed directly.
- **Transaction-level view:** each row is treated as one customer, so results describe transactions rather than changes in individual behaviour.

11.2 Suggested Next Steps

- Add purchase date, channel and margin data to enable trend, channel and profitability analysis.
- Add statistical testing to confirm which differences are significant.
- Extend the dashboard with segment and trend views as new data becomes available.
- Explore predictive models, such as subscription propensity or churn, once time-based data is available.